2017 Ph H2 Q8

Section: Particles and Waves

Topic: The Standard Model

A student is studying the Standard Model of particle physics.

(a)

State the name of the force responsible for holding protons and neutrons together in the nucleus.

✅ The **strong nuclear force**.

> This force acts over very short distances and **overcomes the electrostatic repulsion** between protons.

(b)

The muon and the electron are both classified as leptons.
State one difference between a muon and an electron.

✅ The **muon is more massive** than the electron.

> Another possible answer: the muon is **unstable** and decays, while the electron is stable.

(c)

State the name of the force responsible for the decay of a muon.

✅ The **weak nuclear force** (or **weak interaction**).

> It governs processes like **beta decay** and **muon decay**, where particles change type (flavour).

🧠 Revision Tips:

- **Strong force**: binds nucleons (protons + neutrons)
- **Weak force**: responsible for particle **decay**
- **Muon**: same charge as electron, but **heavier and unstable**
- **Leptons**: include electron, muon, tau (and their neutrinos)
